ExamGOAL
PYQ PDF

Free · With answer key

GATE ECE · CONTROL SYSTEMS · CHAPTER-WISE PYQS

Basic of Control Systems

GATE ECE Control Systems · Previous Year Questions with Answer Key

11	11	1994 – 2020	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	8 questions
Marks 2	3 questions
Answer key	at the end

Practise online

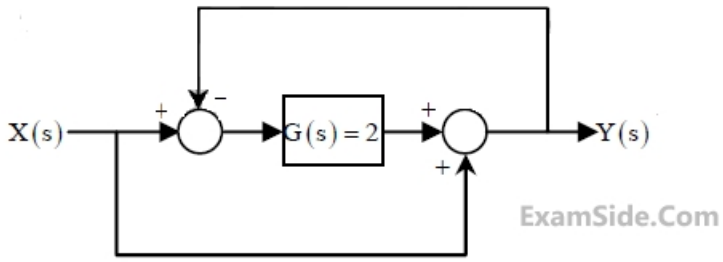
Every question with its full solution, free on **questions.examside.com**

1. **Solve** — Print it, or work through it on screen.
2. **Check** — The answer key is on the last pages.
3. **Understand** — Scan the QR code beside any question for its full solution.

Marks 1 8 questions

Q1. GATE ECE 2017 Set 2

For the system shown in the figure, Y (s) / X (s) = _____.



Open solution

Q2. GATE ECE 2017 Set 1

Consider a stable system with transfer function

$$G(s) = \frac{s^p + b_1 s^{p-1} + \ldots + b_p}{s^q + a_1 s^{q-1} + \ldots + a_q}$$

Where $b_1, \ldots\ldots, b_p$ and $a_1, \ldots\ldots, a_q$ are real valued constants. The slope of the Bode log magnitude curve of G(s) converges to -60 dB/decade as $\omega \to \infty$. A possible pair of values for p and q is

(A) p = 0 and q = 3 (B) p = 1 and q = 7 (C) p = 2 and q = 3 (D) p = 3 and q = 5

Open solution

Q3. GATE ECE 2015 Set 3

The impulse response of an LTI system can be obtained by

(A) differentiating the unit ramp response
(B) differentiating the unit step response
(C) integrating the unit ramp response
(D) integrating the unit step response

Open solution

Q4. GATE ECE 2015 Set 1

Negative feedback in a closed-loop control system DOES NOT

(A) reduce the overall gain
(B) reduce bandwidth
(C) improve disturbance rejection
(D) reduce sensitivity to parameter variation

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q5. GATE ECE 2005

Open solution

Despite the presence of negative feedback, control systems still have problems of instability because the

(A) Components used have nonlinearities.

(B) Dynamic equations of the subsystems are not known exactly.

(C) Mathematical analysis involves approximations.

(D) System has large negative phase angle at high frequencies.

Q6. GATE ECE 1998

Open solution

The transfer function of a tachometer is of the form

(A) Ks

(B) $\frac{K}{s}$

(C) $\frac{K}{(s+1)}$

(D) $\frac{K}{s(s+1)}$

Q7. GATE ECE 1995

Open solution

The transfer function of a linear system is the

(A) Ratio of the output, $V_0(t)$ and input $V_i(t)$

(B) Ratio of the derivatives of the output and the input

(C) Ratio of the Laplace transform of the output and that of the input with all initial conditions zeros

(D) none of these

Q8. GATE ECE 1994

Open solution

Tachometer feedback in a d.c. position control system enhances stability.

(A) TRUE

(B) FALSE

Marks 2 3 questions

Q9. GATE ECE 2020

Open solution

Consider the following closed loop control system



where $G(s) = \frac{1}{s(s+1)}$ and $C(s) = K\frac{s+1}{s+3}$. If the steady state error for a unit ramp input is 0.1 , then the value of K is ____ .

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q10. GATE ECE 2003

Let x(t) be the input to a linear, time-invariant system. The required output is 4x(t-2). The transfer function of the system should be

(A) $4\,e^{j4\pi f}$ (B) $2\,e^{-j8\pi f}$ (C) $4\,e^{-j4\pi f}$ (D) $2\,e^{j8\pi f}$

Open solution

Q11. GATE ECE 2000

A linear time invariant system has an impulse response e^{2t}, $t > 0$. If the initial conditions are zero and the input is e^{3t}, the output for $t > 0$ is

(A) $e^{3t} - e^{2t}$ (B) e^{5t} (C) $e^{3t} + e^{2t}$ (D) None

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

Marks 1

1. **1**	2. **A**	3. **B**	4. **B**	5. **D**
6. **A**	7. **C**	8. **A**		

Marks 2

9. **30**	10. **C**	11. **A**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs